

Glass Core Substrates – Advanced Packaging Pivot driving future AI and HPC Chip Performance

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Abstract

Substrates are the foundation for Heterogeneous Integration that is driving the AI and HPC industry. Future AI and HPC packages demand more chiplets integrated on hyper larger form factor substrates. Glass, with its superior mechanical properties, is an ideal material to replace organic core to address this need. In this paper, we showcase some of the breakthrough technologies and materials innovations Intel foundry has developed to enable this technology.

Author Keywords

Glass Core Substrates; Large Form Factor; Feature Scaling; Bump Pitch Scaling; AI; HPC; Heterogeneous Integration; CTE; Chiplets.

1. Introduction

The ever-increasing demand for more powerful compute and the inevitable move of the semiconductor industry into the heterogeneous era that integrates multiple “chiplets” in a package will require many advances in substrates for packages. The substrate industry has gone through major technology transitions about every 15 years. The most recent major transition happened in the late 90s going from ceramic to organic flip chip substrates and became the standard for the industry, known as Organic Core Substrate. With the dawn of AI era, chip architects are demanding high-density, high-performance packages with increased levels of heterogeneous die integration for data-intensive workloads and the afore mentioned organic core substrates can no longer scale [1-4]. As a result, over the last decade, silicon interposers (Chip on Wafer on Substrate, CoWoS) [5] and Embedded Multi-die Interconnect Bridge (EMIB) have emerged to enable ultra-high-density interconnects [6-9]. These have been able to bridge the critical paths of the chips with fast and dense pieces of silicon, albeit at fairly high costs and without addressing the bottlenecks of organic substrate scaling. Glass core substrates with their superior mechanical, physical, and thermal properties can address all the drawbacks, and hence can be a true replacement for the organic core substrates as well as to enable continued scaling for the next generation of semiconductor products.

The benefits of using glass as a core material for substrates have been known for a while, but the success was limited to small coupon demonstrations. This was largely due to the inability to handle glass at large sizes and to process defect-free glass [10]. Recently, our focused efforts resulted in significant strides in both equipment and technology overcoming these limitations enabling successful fabrication of large layer count thick glass core substrates with fully copper filled Through Glass Vias (TGVs) and Embedded Silicon Bridges (EMIB). In this report, we discuss some of the pioneering technologies and materials innovations involved in enabling the making of these substrates. The results presented are significant advances paving the path for successful commercialization of this technology and enabling the scalability of the packages to finer design rules, ultra large area die complexes and Hyper Large Form Factor demanded by HPC and AI applications.

2. Results

Core Through Hole Formation and Metallization:

Cores used in traditional organic core substrates are made up of glass fiber-epoxy laminates. Through hole density in these organic cores is limited due to Cu electromigration along the fiber resin. Glass being a homogeneous material doesn't suffer from such limitations but historically known traditional glass hole formation methods such as CO₂ (carbon dioxide), excimer, or long-pulse lasers often resulted in cracking and chipping. Recently, Laser-Assisted Etching (LAE) process has been developed that works by using an ultrashort, high-peak laser pulses to induce nonlinear absorption (multiphoton and avalanche ionization), creating a narrow laser-affected zone in which glass network is locally modified by Si-O-Si bond disruption and the formation of non-bridging oxygen and defect sites. These laser-modified regions exhibit significantly enhanced chemical reactivity and etch selectivity, enabling smooth, low-chipping, TGH formation [11, 12].

We optimized the process to repeatedly produce low taper, high Aspect Ratio (AR), and tighter through hole pitches at panel scale. Pitches that can match with chip bumps have been demonstrated which can lead to a 10-fold improvement in the through hole density. The Glass holes formed above were subsequently filled with copper. High AR plating in glass substrates had been one of the bottlenecks in the industry to realize high through hole density at core thicknesses of above 500 µm. Figure 1 below shows the cross-sections and X-ray images of our fully copper-filled TGVs at high AR. No voids or defects are seen.

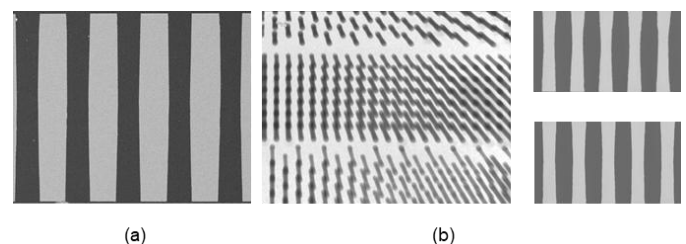


Figure 1. (a) Cross-section of high AR Through Glass Holes formed with LAE process, (b) X-Ray and cross-section data showing void-free, fully copper filled TGVs.

Overlay Improvement:

As discussed earlier, glass has superior mechanical properties than glass fiber-epoxy laminate that makes up the core in the traditional organic core substrates. Glass has higher mechanical/dimensional stability resulting in improved layer-to-layer interconnect overlay, ultra-low flatness enabling a lower depth of focus for improved lithography and can withstand higher temperatures. This can result in as much as 50% less pattern distortion and correctable symmetric shrinkage than organic core materials as shown in figure 2 (a). In addition, the dimensional stability of the core in the horizontal (X) vs vertical (Y) is shown

below in figure 2 (b). All of these taken together can lead to a more stable substrate to mount the dies on that can be interconnected with fine interconnections as can be seen from the figure 2 (c) below. This will open the door to ultra-large form factor packages with very high assembly yields.

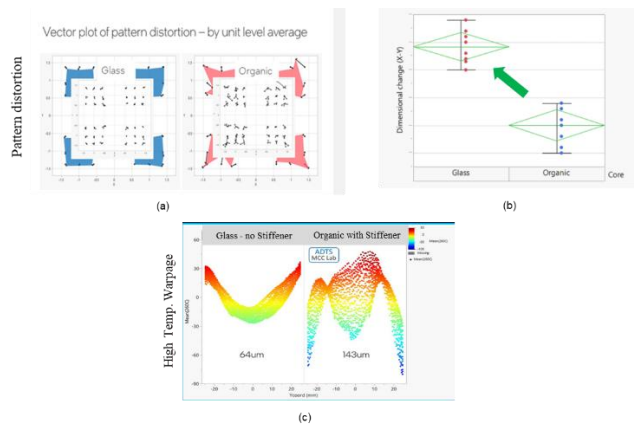


Figure 2. (a) Panel maps showing measured residual overlay error by substrate, (b) Measured dimensional change (X-Y) between process layers, (c) Improved high temperature warpage with glass core substrates as compared to organic core for die attach.

Singulation:

Glass is a strong material; however, it is also brittle. Under tensile stress conditions, flaws and defects, larger than certain thresholds, can lead to crack formation and propagation. In package substrates containing laminated dielectric or Redistribution Layers (RDLs) present on both sides, this manifests itself as failure known as “SeWaRe” during singulation [13]. Intel Foundry Services developed a dicing technology and optimized it to demonstrate 24-layer, thick glass core substrates cut to units with no SeWaRe or any other defects (Figure 3 (a) & (b)). The units were also subjected to the rigorous thermal shock treatment TSC50, and no degradation was seen (Figure 3 (c)).

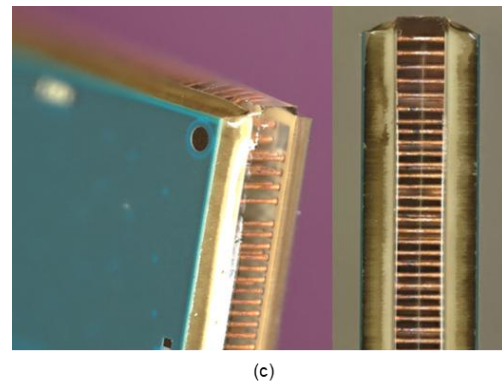
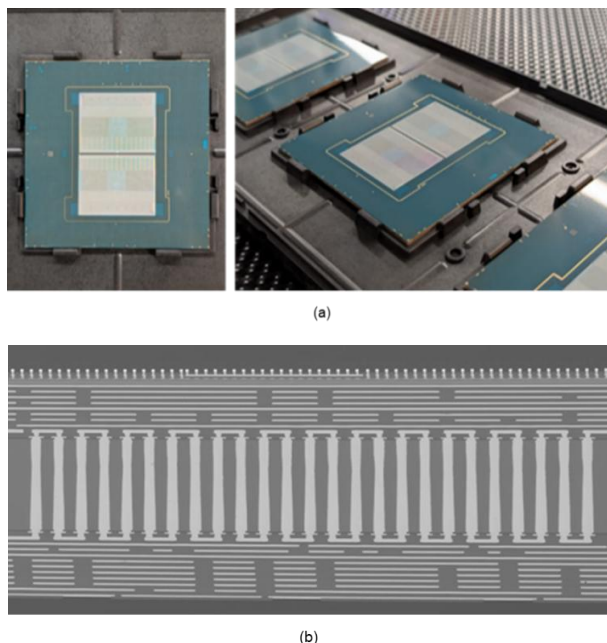


Figure 3. (a) Singulated 24-layer Glass Core Substrate with two embedded bridges with no SeWaRe, (b) Cross Section image of substrate in (a), (c) Cut unit showing no degradation post thermal shock cycling.

3. Conclusions

Scaling of Organic Core substrates is running out of gas due to intrinsic limitations in dimensional stability, flatness / thickness uniformity and through-hole density. Glass core substrates with their superior mechanical, physical, and thermal properties can address these drawbacks, and replace organic substrates to enable continued scaling for the next generation of semiconductor products. Although, the benefits of using glass as a core material for substrates have been known for a while, inability to handle glass at large sizes and to process defect-free glass have limited demonstrations to small coupons. Significant strides in both equipment and technology driven by Intel Foundry Services recently resulted in the industry first demonstration of large layer count thick glass core substrates manufactured at panel scale and paving the path for successful commercialization in the near future.

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